Status	Finished
Started	Wednesday, 21 October 2026, 12:04 PM
Completed	Wednesday, 21 October 2026, 12:06 PM
Duration	2 mins 50 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Duplicates
- c. All of the above
- d. Selection

The correct answer is: All of the above

Question **2**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It requires external libraries
- c. It only works with integer arrays
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is difficult to implement
b. Bubble Sort is less efficient for large lists
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort does not guarantee sorted order

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort
b. Linear Search
c. Quick Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It bubbles up the largest element in each pass
c. It recursively sorts subproblems
d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **7**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Searching

b. Merging

c. Sorting

d. Complexity

The correct answer is:
Complexity

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Searching for data in a list

b. Deleting data from a list

c. Inserting data into a list

d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search

b. Merge Sort

c. Bubble Sort

d. Binary Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Combining sorted sublists
b. Comparing each element with the others
c. Dividing the input into smaller subproblems
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Heap sort
b. Merge sort
c. Selection sort
d. Quick sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. Each part is searched for a specific element
c. The parts are merged without sorting
d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It allows for efficient parallel processing and sorting of data
c. It avoids the need for recursion
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **14**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Selection
c. Distribution
d. Insertion

The correct answer is:
Insertion

Question **15**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of non-duplicate keys
b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys